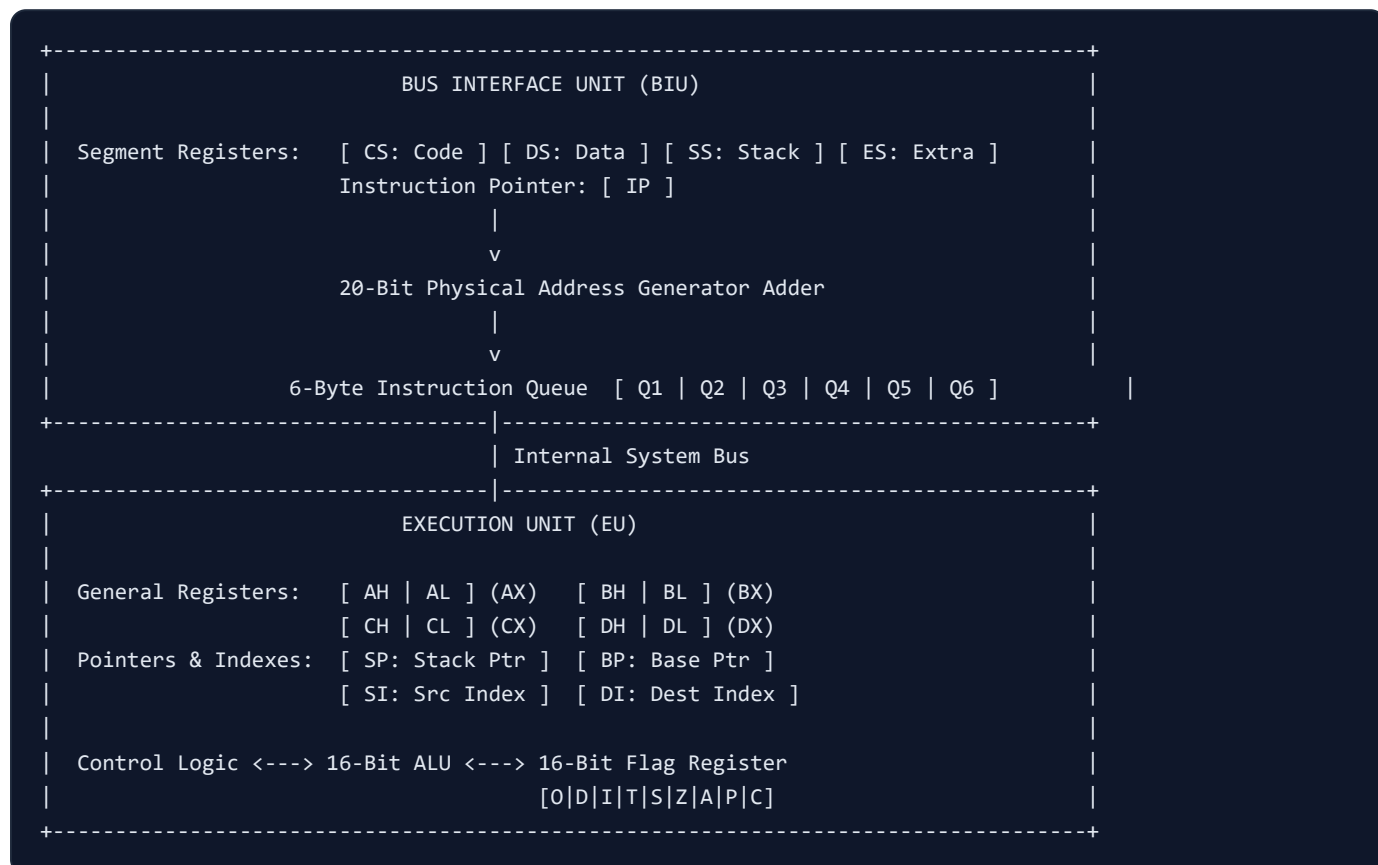


10-Mark Questions & Answers — MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Explain the internal architecture of 8086 in detail (BIU, EU, registers, flag register) with a neat labeled diagram.



1. Bus Interface Unit (BIU)

- **Segment Registers:** Four 16-bit registers (CS, DS, SS, ES) storing segment base addresses.
- **Instruction Pointer (IP):** Holds the 16-bit offset of the next instruction byte to fetch.
- **Instruction Queue:** 6-byte FIFO buffer prefetching instructions during idle memory cycles.

2. Execution Unit (EU)

- **ALU (16-bit):** Performs arithmetic and bitwise logic operations.
- **General Purpose Registers:** AX (Accumulator), BX (Base), CX (Counter), DX (Data).
- **Pointer/Index Registers:** SP, BP, SI, DI.
- **16-bit Flag Register (9 Active Flags):**
 - **Status Flags:** CF (Carry), PF (Parity), AF (Aux Carry), ZF (Zero), SF (Sign), OF (Overflow).
 - **Control Flags:** IF (Interrupt Enable), DF (Direction Flag), TF (Trap Flag).

Q2. Explain the functional pin diagram of 8086 microprocessor in detail, describing the function of each pin group (address/data bus, control signals, power supply).

	+---	---	+
GND	1	40	VCC (+5V)
AD14 <->	2	39	<-> AD15
AD13 <->	3	38	-> A16/S3
AD12 <->	4	37	-> A17/S4
AD11 <->	5	36	-> A18/S5
AD10 <->	6	35	-> A19/S6
AD9 <->	7	34	-> BHE' /S7
AD8 <->	8	33	-> MN/MX'
AD7 <->	9	32	-> RD'
AD6 <->	10	31	-> HOLD (RQ/GT0')
AD5 <->	11	30	-> HLDA (RQ/GT1')
AD4 <->	12	29	-> WR' (LOCK')
AD3 <->	13	28	-> M/IO' (S2')
AD2 <->	14	27	-> DT/R' (S1')
AD1 <->	15	26	-> DEN' (S0')
AD0 <->	16	25	-> ALE (QS0)
NMI -->	17	24	-> INTA' (QS1)
INTR -->	18	23	<- TEST'
CLK -->	19	22	<- READY
GND	20	21	--> RESET
	+-----+		

- 1. Multiplexed Address/Data Bus (AD0 - AD15):** Time-multiplexed lines carrying lower 16-bit memory address during **T1** and 16-bit data during **T2-T4**.
- 2. Address/Status Lines (A16/S3 - A19/S6):** Upper 4-bit address lines multiplexed with CPU status indicators.
- 3. Minimum Mode Control Signals:**
 - **ALE** (Address Latch Enable), **RD'**, **WR'**, **M/IO'**, **DT/R'**, **DEN'**, **INTA'**.
- 4. Maximum Mode Status Lines:**
 - **S0', S1', S2'** decoded by 8288 bus controller into memory/IO read-write commands.
- 5. System Controls & Power:** **VCC** (+5V), **GND**, **CLK** (Clock input 5-10MHz), **RESET**, **READY**, **NMI**, **INTR**.

Q3. Explain memory segmentation and banking in 8086 in detail with physical address calculation examples. Also explain why segmentation is used.

1. Why Memory Segmentation is Used

- Allows a 16-bit CPU architecture with 16-bit registers to address **1MB (2²⁰ bytes)** of physical memory.
- Enables modular code relocation and dynamic memory protection between code, data, and stack spaces.

2. Physical Address Calculation Formula & Example

Physical Address = (Segment Register × 16) + Offset

- **Example 1 (Instruction Fetch):** **CS = 3000H**, **IP = 0100H**
Physical Address = (3000H × 16) + 0100H = 30000H + 0100H = 30100H

- **Example 2 (Data Read):** $DS = 4500H$, $BX = 0250H$

$$\text{Physical Address} = (4500H \times 16) + 0250H = 45000H + 0250H = 45250H$$

3. Memory Banking System (Even & Odd Banks)

1MB PHYSICAL MEMORY SYSTEM		
/	EVEN BANK (512 KB)	ODD BANK (512 KB)
Address Line A0 = 0	Bus High Enable BHE' = 0	
Data Bus D0 - D7	Data Bus D8 - D15	

Q4. Explain the functioning of 8086 in both minimum mode and maximum mode with neat diagrams, and state the key differences.

1. Minimum Mode Block Diagram (MN/MX' = 1)

```

+-----+ ALE +-----+ -> 20-bit Address Bus
| 8086 CPU | ----->| 74LS373 |
| (MN/MX=1) | +-----+
| | AD0-15+-----+ -> 16-bit Data Bus
| | ----->| 74LS245 |
| | +-----+
| | Control+-----+ -> Memory/IO Controls (RD, WR, M/IO)
+-----+ ----->| Memory/IO |
                +-----+

```

2. Maximum Mode Block Diagram (MN/MX' = 0)

```

+-----+ S0',S1',S2' +-----+ -> Control Signals (MRDC, MWTC, IORC, IOWC)
| 8086 CPU | ----->| 8288 BUS |
| (MN/MX=0) | | CONTROLLER |
+-----+ +-----+

```

Q5. Draw and explain in detail the timing diagrams for read and write operations in both minimum and maximum mode of 8086.

MINIMUM MODE READ TIMING DIAGRAM:

```

CLK      :  _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
          | T1 | T2 | T3 | T4 |
ALE      :  _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
A16-A19  :  == Address == Status ==
AD0-AD15 :  == Address == High-Z == Data ==
RD       :  _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

```

Q6. Explain the interrupt structure of 8086 in detail, including types of interrupts (hardware/software), interrupt vector table, and interrupt servicing procedure.

```
INTERRUPT VECTOR TABLE (00000H - 003FFH, 1KB RAM)
+-----+ 003FFH
| Vector 255 (Type 255) |
| ... |
| Vector 2 (NMI - Non-Maskable Pin) |
| Vector 1 (Single Step / Trap) |
| Vector 0 (Divide-by-Zero Error) |
+-----+ 00000H
```

Q7. Explain the process of demultiplexing the address/data bus in 8086 with a circuit diagram (using latch 74LS373).

